HPC_DEMO

Class-Scale Parallel Optimization Study + N=200 Hero Run

Non-Convex Polygon Packing Under Fixed Resource Constraints

Joel Amir Dario Maldonado Tänori

April 8, 2026

Contents

Chapter 1

Introduction

1.1 Purpose of the Study

This work evaluates three parallel optimization strategies for minimizing the side length L of a square containing N identical non-convex polygons.

This is a **demonstrative algorithms study** under strict class-project constraints:

- Maximum 20 CPU cores
- ≤ 4 hours wall-clock per method
- One N=200 hero run
- Clear, interpretable convergence plots

1.2 Study Structure

We split the evaluation into:

- **Graph Suite:** $N \in \{5, 10, 20, 50, 100\}$
- **Hero Suite:** $N = 200$

The hero run is a demonstration of capability, not a statistical comparison.

Chapter 2

Problem Formulation

2.1 Feasibility Definition

A configuration is feasible if:

$$\texttt{outside\_penalty} \leq \texttt{EPS\_FEAS} \quad \text{and} \quad \texttt{overlap\_penalty} \leq \texttt{EPS\_FEAS}$$

2.2 Primary Metric

$$L_{\text{best}}(t) = \min_{\tau \leq t} \{L(\tau) : \text{feasible at time } \tau\}$$

2.3 Secondary Metrics

- Final $L_{\text{best}}(T)$
- Final bracket width $L_{hi} - L_{lo}$
- Success rate
- Minimum penalty E_{min}

2.4 Hero Efficiency Metric

$$\eta = \frac{N A_{poly}}{L^2}$$

Chapter 3

Experimental Design

3.1 Hardware Constraints

- 20 CPU cores maximum

3.2 Parallel Layout

3.2.1 Graph Suite

- $R = 10$ threads per run
- Two seeds concurrently

3.2.2 Hero Run

- $R = 20$
- Single run

3.3 Determinism

$$\texttt{seed\_r} = \texttt{splitmix64}(\texttt{global\_seed} \oplus \texttt{hash}(N, run\_id, probe\_idx, r))$$

Chapter 4

Algorithmic Methods

4.1 Method A: Multi-Start (MS)

Independent simulated annealing workers.

Early stop if any worker becomes feasible.

4.2 Method B: Elite Resample Multi-Start (ER-MS)

Resample frequency:

$$K_{resample} = 200N$$

Elite pool:

$$k' = \lceil 0.25R \rceil$$

Noise schedule:

$$\sigma_{pos} = (0.02 - 0.015f)L$$

$$\sigma_{\theta} = (5^{\circ} - 4^{\circ}f)$$

4.3 Method C: Parallel Tempering (PT)

Geometric temperature ladder.

Swap frequency:

$$K_{swap} = 200N$$

Early stop only if coldest replica feasible.

Chapter 5

Bisection and Scheduling

5.1 Bracket Initialization

$$L_{lo} = \sqrt{NA_{poly}}$$
$$L_{hi} = \gamma(N)L_{lo}$$

5.2 Graph Suite Schedule

Base slice times:

N	t_{base}
5,10,20	6s
50	10s
100	15s

Stop when:

$$L_{hi} - L_{lo} \leq 10^{-3}L_{hi}$$

5.3 Hero Schedule

- 60s slices (probes 1–6)
- 120s slices (7–12)
- 240s slices (13+)

Chapter 6

Results — Graph Suite

6.1 Convergence Plots

6.2 Bracket Evolution

6.3 Success Rates

Insert summary table here.

6.4 Discussion

Discuss coordination effects, scaling behavior, and feasibility trends.

Chapter 7

Results — N=200 Hero Run

7.1 Run Configuration

Describe chosen method and pilot behavior.

7.2 Convergence

7.3 Final Metrics

Report L, bracket width, and efficiency η.

7.4 Final Packing

7.5 Qualitative Analysis

Discuss structural characteristics of the packing.

Chapter 8

Limitations

- Different R values between suites
- Small seed counts
- Hero run is demonstrative only

Chapter 9

Reproducibility and Artifacts

- Commit hash
- Config files
- Seeds
- Output CSV files
- SVG files